Laboratory Transition and Experimental Protocol for the AGT Program: Resonance, Screening, and Bubble-Strip Bench Tests

Companion operational manuscript

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Abstract

This manuscript operationalizes the first hardware-facing layer of the AGT program in bench and measurement language. It is downstream of the AGT theory manuscript and the AGT computational manuscript rather than a proof document, a phenomenology fit, or a finished device paper.

The companion translates the AGT observable dictionary into strip-resonance, bubble-capture, interface, and mixed-medium bench protocols. It defines laboratory surrogates for signed grav response, spectral branch compatibility, mesoscopic field reconstruction, transport fractions, and large-scale distinction criteria. The present release is limited to proof-of-regime bench work and does not claim theorem closure, cosmological fit, or a macroscopic engineering antigravity device.

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1 Protocol, Boundary, and Laboratory Status Policy

1.1 Purpose of the laboratory companion

This companion translates the AGT verification ladder into apparatus-level practice. Its role is to define the first disciplined bridge from source-faithful numerics and emulator artifacts to resonant, transport, and mixed-medium bench campaigns.

1.2 Boundary of the manuscript

The present document is downstream to the AGT theory manuscript and the AGT computational manuscript. It is not theorem evidence for AG-T1 through AG-T16, not a completed phenomenological closure, and not a claim that large-scale engineering antigravity has already been demonstrated.

Shared boundary clause. This companion is downstream to the AGT theory manuscript; it uses the theory vocabulary without upgrading bench data into theorem evidence; and it operationalizes only the first proof-of-regime hardware layer.

1.3 Laboratory status labels

The laboratory companion uses a dedicated status system:

- [LA] Laboratory Assumption.
- [LD] Laboratory Definition.
- [LP] Laboratory Protocol.

- [LM] Measured Observable.
- [LR] Laboratory Result.
- [LF] Laboratory Failure Mode.
- [LT] Laboratory Target Claim.

2 Translation Map from Theory to Bench Practice

2.1 Theory objects carried into the bench layer

Theory object	**Bench counterpart**	**Measured or controlled channel**	**What is not claimed**
q_g	Signed grav-response witness	signed resonance or transport contrast	no observer-independent microscopic grav charge
$Q_{g,\mathrm{int}}^{\mathrm{spec}}$	branch-compatible spectral witness	peak position, peak sign, and branch separation	no exact inverse reconstruction of the microscopic source
$\rho_g, \phi_g, E^{(g)}, F^{(g)}$	mesoscopic reconstructed field proxies	reconstructed density, potential, field, and effective-force channels	no unique coarse-graining prescription
capture, reflection, transmission, trapping	bench transport fractions	strip and interface transport logs	no universal transport law outside the frozen bench family
wall, shell, void, bubble variables	structured-medium observables	bubble radius, shell response, interface contrast, mixed-medium distinction	no cosmological or geophysical fit
falsifier ladder	bench stop rules	failed witness, collapsed window, neutralization, or indistinguishability flags	no rhetorical rescue of failed runs

2.2 Proof-of-regime baseline

The present laboratory layer freezes three first-release families: a strip-resonance family for repulsive-window extraction, a bubble-cell family for capture and screening proxies, and a mixed-medium distinction family for separating AGT-like signatures from ordinary attractive media. The P3 emulator line remains a pre-screening and comparison package rather than direct evidence.

3 Laboratory Standing Assumptions

3.1 Bench-family assumptions

Assumption 3.1 [LA]. Frozen first-release bench families. The first AGT hardware release is restricted to strip, bubble, and mixed-medium bench families that admit a frozen excitation chain, a stable detector chain, and a reproducible control schedule. Cosmological, geophysical, and free-flight claims are excluded from this release.

3.2 Observable discipline

The AGT verification ladder is preserved in the laboratory order: microscopic or source-facing witness, spectral sign compatibility, mesoscopic field reconstruction, transport discrimination, and only then large-scale distinction. A later layer may not be claimed when an earlier layer has failed.

3.3 Control assumptions

The bench program assumes that excitation frequency, amplitude, medium composition, geometry, and environmental isolation are logged explicitly for each run. Post hoc tuning is not allowed to redefine success after the run is complete.

3.4 Non-claim assumptions

The current release assumes that all bench observables are surrogates. Positive signatures are interpreted only as source-faithful evidence for a restricted AGT-like regime under a frozen measurement protocol, not as unrestricted proof of antigravity in the ordinary-language sense.

4 Bench Observables and Verdict Ladder

4.1 Signed response observable

Definition 4.1 [LM]. Signed response witness. The first bench observable is the signed response witness

$$W_{\mathrm{sgn}} = S_+ - S_-,$$

where S_+ and S_- are the detector-normalized positive and negative branch responses extracted under the frozen excitation and normalization protocol.

A valid AGT bench run must first show that the signed response is nontrivial and reproducibly separated from the null response of the matched control medium.

4.2 Spectral compatibility observable

The second bench observable is the branch-compatibility package

$$W_{\mathrm{spec}} = (\omega_\star, \Delta_{\mathrm{branch}}, \sigma_{\mathrm{sgn}}),$$

consisting of the selected resonance or transport peak $\omega_\star$, the local branch separation Δ_{branch}, and the sign-consistency flag σ_{sgn}. A bench run fails at this layer if the selected response cannot be attached to a stable compatible branch.

4.3 Mesoscopic field proxy

The third bench layer reconstructs the mesoscopic proxy tuple

$$W_{\mathrm{field}} = (\rho_g^{\mathrm{bench}}, \phi_g^{\mathrm{bench}}, E_g^{\mathrm{bench}}, F_g^{\mathrm{bench}})$$

from the frozen inversion or coarse-reconstruction procedure. This is a disciplined proxy layer, not a claim that the laboratory has directly measured observer-independent gravitational source variables.

4.4 Transport and capture observables

The transport layer records strip and interface fractions

$$W_{\mathrm{tr}} = (T_{\mathrm{rep}}, R_{\mathrm{int}}, C_{\mathrm{cap}}, \tau_{\mathrm{trap}}),$$

where T_{rep} is the repulsive-window transmission or pass fraction, R_{int} is the interface reflection proxy, C_{cap} is the bubble-capture or absorption proxy, and τ_{trap} is the trapped-state dwell time when present.

4.5 Large-scale distinction observable

Definition 4.2 [LM]. Mixed-medium distinction observable. The large-scale distinction observable is the signed separation

$$\Delta_{\mathrm{med}} = O_{\mathrm{AGT}} - O_{\mathrm{attr}},$$

between the frozen AGT candidate medium and the matched one-component attractive control medium under the same geometry and control schedule.

The AGT line does not pass the final proof-of-regime gate if Δ_{med} remains unresolved under repeated matched runs.

4.6 Verdict ladder

The hardware verdict ladder is:

- null response;
- sign-only witness without branch compatibility;
- branch-compatible witness without transport discrimination;
- transport-supported local regime;
- mixed-medium distinction success.

Every later verdict class presupposes the earlier ones.

5 Apparatus Architecture

5.1 Strip-resonance bench

This bench family is the first route to a repulsive-window witness. It uses a frozen strip geometry, programmable excitation chain, matched control strip, and detector stack sufficient to extract resonance position, sign, linewidth, and transport fractions.

5.2 Bubble and cavity bench

This bench family targets capture, screening, and trapped-state proxies. The active observables are bubble radius, shell or interface response, absorption or capture contrast, and hold-time stability under a fixed excitation envelope.

5.3 Mixed-medium distinction bench

This bench family compares an AGT candidate medium against an ordinary attractive control medium while preserving geometry, control schedule, and detector calibration. Its purpose is to block the interpretation that every positive-looking signature is only a trivial medium artifact.

5.4 Digital-to-analog comparison layer

The P3 emulator line is used as a pre-screening and comparison package for run-mode selection, sign expectations, and qualitative phase labels. It does not certify theorem closure or replace direct bench observables.

6 Experimental Protocol Families

6.1 Strip-resonance protocol

Protocol 6.1 [LP]. Strip-resonance protocol. Each strip run freezes geometry, drive family, matched control strip, detector gain, and peak-extraction rule before the excitation sweep begins. The minimum output package is a signed response record, a branch-compatibility record, and a transport-fraction record.

The release gate at this layer is a reproducible non-null signed response together with a stable compatible branch and a non-collapsed repulsive-window witness.

6.2 Bubble-capture protocol

Protocol 6.2 [LP]. Bubble-capture protocol. Each bubble run freezes cavity geometry, shell or interface composition, drive schedule, and dwell-time measurement rule. The minimum output package is a capture proxy, an interface response trace, and a trapped-state or collapse classification.

This protocol is the first direct hardware consumer of the computational bubble campaign language.

6.3 Mixed-medium distinction protocol

The mixed-medium distinction run compares candidate and control media under the same excitation and detector chain. A valid distinction run must preserve blinded labeling until after the first-pass verdict has been assigned.

6.4 Run repetition rule

No release-grade claim is made from a single run. Each condition must be repeated three times with the same geometry, control schedule, and calibration state. Any sign flip, branch reassignment, or loss of mixed-medium distinction across repetitions downgrades the condition to unresolved.

7 Acceptance Criteria and Failure Modes

7.1 Acceptance gate

Target Claim 7.1 [LT]. First AGT proof-of-regime target. *The first AGT hardware target is reached only when a bench family exhibits a nontrivial signed response, stable branch compatibility,*

non-collapsed transport discrimination, and a reproducible distinction from the matched attractive control medium.

At the current release stage this is a proof-of-regime criterion, not a macroscopic device criterion.

7.2 Failure modes

Failure Mode 7.2 [LF]. Trivial-sector collapse. *The signed response collapses to a null or control-indistinguishable sector.*

Failure Mode 7.3 [LF]. Branch-incompatibility failure. *The selected response cannot be attached to a stable compatible branch or loses branch separation under repeated runs.*

Failure Mode 7.4 [LF]. Repulsive-window collapse. *The strip or interface transport layer never stabilizes a reproducible repulsive-window witness.*

Failure Mode 7.5 [LF]. Neutralization or screening washout. *The bubble or mixed-medium layer washes out every signed or transport-level distinction before a stable bench verdict can be assigned.*

Failure Mode 7.6 [LF]. Ordinary-medium indistinguishability. *The candidate medium remains operationally indistinguishable from the matched attractive control medium.*

8 Reproducibility, Audit Trail, and Forward Dependencies

8.1 Minimum archival products

Every release-grade condition archives: geometry sheet, medium composition record, detector calibration state, raw trace bundle, peak-extraction record, verdict sheet, and matched-control comparison sheet.

8.2 Operational file crosswalk

The first AGT hardware release is backed by a frozen markdown asset bundle:

- campaign protocol: `hardware/protocols/agt_p3_campaign_protocol.md`;
- run sheet: `hardware/protocols/agt_p3_run_sheet_template.md`;
- baseline BOM: `hardware/bom/agt_p3_baseline_bom.md`;
- calibration sheet: `hardware/calibration/agt_p3_calibration_sheet.md`;
- anchor library: `hardware/calibration/agt_p3_anchor_library.md`;
- geometry note: `hardware/drawings/agt_p3_strip_bubble_geometry_spec.md`.

8.3 Dependency order

The forward dependency order of the hardware line is

$$\text{signed response} \to \text{branch compatibility} \to \text{field proxy} \to \text{transport discrimination} \to \text{mixed-medium distinction} \to \text{larger prototype claims}.$$

This order is strict: a later layer may not be used to repair a failed earlier layer.

8.4 What is not claimed

The present laboratory companion does not claim complete theorem closure of AG-T1 through AG-T16, a finished cosmological or phenomenological fit, a macroscopic engineering antigravity platform, or observer-independent antigravity outside an explicitly frozen regime window.